Mechanical performance of fiber-reinforced concrete with coconut coir as self-healing agent

Author links open overlay panelHalden Clyde P. Manuta [a c 1], Julia Lycci V. Martos [a 1], Harley Julien P. Dillo [a 1], Roumel Salvador Alvarez [a c], Jazth D. Manota [b c], Crijamaica L. Oceña [b c], Noel Peter B. Tan [d], Chosel P. Lawagon [b c]

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Highlights

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Coconut coir fibers modified with silica for self-healing concrete.

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Self-healing coir (1–5% replacement) improved concrete compressive strength by 18%.

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5% self-healing coir composite in concrete enabled rapid crack healing by day 7.

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SEM-EDX confirmed $CaCO_3$ precipitation as the main healing mechanism.

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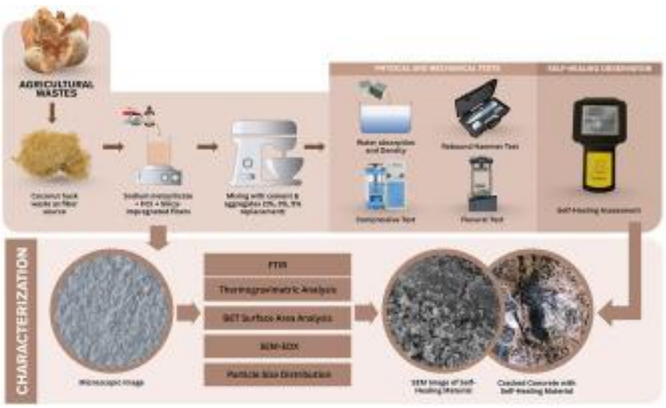
Coconut waste was valorized for sustainable and resilient construction.

Abstract

Concrete is commonly used in construction because of its high compressive strength and durability. However, its brittle nature makes it vulnerable to microcracks that can decrease long-term performance. This study evaluates coconut coir fiber (CCF) as a sustainable additive for self-healing concrete. CCF was incorporated as a partial cement replacement (1%, 3%, and 5% by weight), in both its natural form and as a carrier for a silica-based self-healing material (SHM). Physical and mechanical properties, including density, water absorption, compressive strength, and flexural strength, were examined together with crack-healing efficiency. The results showed that CCF exhibited high tensile strength

(~115.5 MPa). Specimens with SHM-CCF at 3% and 5% showed significant reductions in crack width after 28 days (about 75-80%), indicating enhanced self-healing ability. Although water absorption slightly increased with higher fiber content, concrete densities stayed within the normal range (2230-2290 kg m^{-3}). The highest flexural strength (~5.15 MPa) was reached at 1% SHM-CCF, while higher amounts caused reduced strength due to fiber agglomeration. Overall, SHM-CCF shows potential as a sustainable cementitious additive that improves crack healing and mechanical performance while utilizing agricultural waste.

Graphical abstract



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Introduction

Concrete has been extensively used as a primary structural material throughout the 20th and 21st centuries due to its wide availability, high compressive strength, and adaptability to diverse construction applications. Despite continuous advances in concrete technology, new challenges have emerged, most notably the increasing susceptibility of concrete to cracking [1,2]. Cracks may develop not only under applied loads but also from drying shrinkage, thermal stresses, and chemical reactions, creating pathways for aggressive agents that accelerate deterioration of the concrete matrix over time [1]. This concern becomes even more critical in seismic regions such as the Philippines, which is located along the Pacific Ring of Fire and ranks among the world's most disaster-prone countries with a disaster risk index of 25.14% [3]. Recent consecutive high-magnitude earthquakes in Mindanao have highlighted the vulnerability of concrete infrastructure, especially aging structures that no longer comply with updated seismic design standards [4]. While conventional repair techniques exist, microcracks often go undetected and gradually compromise durability, and manual repair interventions are both costly and inefficient [5].

These challenges underscore the growing need for self-healing concrete systems that can autonomously control crack propagation and extend the service life of concrete structures [1].

Material-based additives have gained increasing attention for their ability to enhance construction practices by enabling reinforcement and intrinsic repairing mechanisms, thereby reducing the maintenance costs of concrete structures. This is done by incorporation of micro-capsules [6,7], crystalline admixtures [6,8], microbial carbonate precipitation [9] (MCP) [10], electrodeposition [11], and engineered-cementitious composites (ECC) [10]. These approaches primarily function by promoting secondary hydration reactions, wherein calcium hydroxide ($Ca(OH)_2$) leaches from the matrix and reacts with water and available carbon dioxide (CO_2) to form calcium carbonate ($CaCO_3$) crystals that seal cracks [11,12]. When this process occurs naturally, it is classified as autogenous healing, whereas deliberate enhancement through engineered additives or biological agents constitutes autonomous self-healing [12,13]. While these strategies highlight the growing emphasis on integrating material innovation with intrinsic healing mechanisms to improve concrete durability, their effectiveness remains highly dependent on material characteristics, crack geometry, and environmental conditions.

Accordingly, establishing clear performance limitations and evaluation metrics is essential when assessing the efficiency of self-healing materials (SHM) [13]. Previous studies have reported crack closure rate of Portland cement-based composite can seal the cracks at rate of 15 μm day-1 [14], while MCP at 0.5~5 μm day-1 [15]. The extent of crack closure observed in the literature varies widely, with reported crack widths ranging from 50 to 800 μm, where complete healing is typically achieved for cracks up to ~0.4-0.5 mm, and only partial closure is observed for wider cracks of 0.7-0.8 mm under favorable conditions [14]. Within this context, silica (SiO_2) in various forms has been incorporated either as a supplementary additive or in combination with other self-healing agents [6,[16], [17], [18]]. At low dosage (0.2-1.0%), silica nanoparticles react with portlandite and produce an excess calcium-silicate-hydrate (CSH) resulting in early strength development [16,17,19], pore refining effect, and improvement of interfacial transition zone [17,20] creating a more durable and resilient concrete. Furthermore, SiO_2 precursors such as sodium silicate solutions (SS), in both soluble and insoluble form, are commonly incorporated through polymer-based encapsulation to produce micro- or nano-capsules, with healing performance strongly dependent on the polymer shell-to-core ratio [6,21]. Some studies have employed glass encapsulation [22,23]; however, this approach often suffers from limited chemical activation due to the rigidity and geometry of the glass shell. These approaches have demonstrated notable enhancements in healing efficiency; however, several limitations persist. Reductions in compressive strength have been reported, primarily attributed to inadequate interfacial bonding between the microcapsules and the cement matrix, heterogeneity in shell thickness, and the surface roughness of the encapsulating layer. Collectively, these factors impair the rupture efficiency of the capsules, thereby constraining their chemical reactivity and overall performance within the cementitious system [9,20,24].

In parallel with the pursuit of self-healing technologies, material-based approaches aimed at reducing construction and maintenance costs of concrete structures have gained increasing attention. Among these, the incorporation of natural fibers (NF) (abundant byproducts of agricultural processes) has been extensively studied due to renewability, availability, and compatibility with cementitious systems [[25], [26], [27]]. Previous investigations indicate that the addition of NF at levels of ≤ 1-2 % [27] by weight of cement replacement can enhance compressive and flexural capacity of ∼15-25% [28,29] largely due to their crack-bridging capability [30]. Furthermore, the water-retaining capacity of these hydrophilic materials allows for gradual water release, which serves as nucleation sites, hence, increases the production of hydration products [25,31]. This approach is particularly relevant in the Philippine context, where coconut production accounts for nearly a quarter of global output, yet 70% of coconut biomass is discarded as waste [32,33]. Coconut coir fiber (CCF) emerges as a promising alternative owing to its high lignin (40-45%) and cellulose (32-44%) content, moisture content (~8%), favorable tensile strength (65-230 MPa), and high strain capacity (15-24%), which collectively contribute to rigidity and toughening effect [[34], [35], [36]] Studies on natural fiber reinforced concrete (NFRC) with CCF resulted in an increase of roughly 19–20% increases in both 28-day compressive and flexural strength of concrete [34]. Noticeably, because of its fibrous structure, previous studies also observed that a decrease in workability slump was reduced by over 70% due to factors involving length-to-diameter ratio, dry or pre-wetting preparation of fiber, and dispersion [35]. Moreover, morphology obtained from untreated CCF revealed a smooth texture which hinders its binding efficiency. Hence, incorporation of chemical treatments like alkali treatment, acetylation, and silanization is employed to increase its interfacial bonding and reduce hydrophilicity, leading to a higher strength capacity [27,36,37].

Despite these advances, many sophisticated self-healing technologies remain economically and practically constrained due to energy-intensive extraction processes, reliance on encapsulation technology to ensure stability, and limited secondary hydration caused by insufficient nucleation sites for effective crack sealing [24]. Addressing these challenges, this study introduces a new and underexplored concept by using porous coconut (*Cocos nucifera*) coir fiber as a substrate for the sol–gel precipitation of silica nanoparticles. This strategy enables the coir fiber to function simultaneously as a natural reinforcement and as a carrier of silica-based SHM within natural fiber-reinforced concrete as cement replacement. The investigation focuses on the synthesis and characterization of the resulting composite, its crack-sealing capability as reflected by visual healing and crack width reduction over time, and its influence on key physical and mechanical properties, including water absorption, density, and compressive and flexural performance assessed through destructive and non-destructive methods. The combined use of coconut coir fiber and silica is further expected to produce a synergistic effect by enhancing hydration kinetics and contributing to improved overall concrete performance.

Materials

Coconut husks were collected from Toril, Davao City, Philippines. The self-healing material was synthesized using sodium metasilicate (Na_2SiO_3 AR, Neutral Solution, ERN Chemical Industry) and hydrochloric acid (36–38% HCl AR, Macron). Necessary construction materials were obtained locally, including fine aggregates, coarse aggregates (Baure Sand & Gravel), and blended cement (Type 1T-MH, Holcim Excel ECOPlanet).

Preparation of raw materials

Coconut husks were retted, chopped, and manually extracted to separate fibrous and

Physical properties of CA and FA

The characterization of CA and FA (Table 2) shows that FA displays a consistent gravimetric density with moderate porosity, as indicated by relative and apparent relative densities. The significant water absorption capacity of FA suggests its potential for absorbing water. In comparison, CA exhibits lower water absorption potential, and its specified bulk density implies a higher mass per unit volume in its natural state. CA displays higher relative and apparent densities, indicative of lower

Conclusions

This study demonstrated that SHM-CCF enhances compressive strength and promotes effective crack healing through the precipitation of $CaCO_3$. While natural coir fibers alone reduced compressive strength at higher loadings, SHM modification improved bonding and hydration, mitigating this drawback. Flexural strength benefits were evident at low SHM dosages but declined with higher content due to fiber clustering, highlighting the need for optimized dispersion strategies. The study was limited to

CRediT authorship contribution statement

Halden Clyde P. Manuta: Writing – review & editing, Visualization, Project administration, Methodology, Investigation. **Julia Lycci V. Martos:** Writing – review & editing, Writing –

original draft, Visualization, Validation, Investigation. **Harley Julien P. Dillo:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation. **Roumel Salvador Alvarez:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Jazth D.**

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:Roumel Salvador Alvarez has patent #PH22025050169 pending to n/a. Given his/her/their role as Editor, had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that

Acknowledgment

This study was funded by the Department of Science and Technology - Philippine Council for Industry, Energy, and Emerging Technology Research and Development (DOST-PCIEERD) through the Center of Green Nanotechnology Innovations for Environmental Solutions (CGNIES).